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Never Type It Twice

Links and Syncs between Apps on Desktop and in the Cloud

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Introduction

Overview

Never type it twice. You and your firm can minimize the wasted time and hassles of reentering the same information and text in multiple places.

You cannot keep all of your contact, calendar, and case information in one place in a single computer program or Cloud application. You need to reuse that information over and over again in your communications and records.

Client names need to be inserted into documents, Emails, phone messages, tasks, invoices and reports. Important dates need to appear on your calendar, in documents, in messages and reminders. You need to see them on different devices such as your office computer screen, your smartphone, your tablet, your laptop and your home computer.

The challenge is to come up with a combination of applications, services and devices that works efficiently for you and your organization. Whether you are in a solo practice, a law firm or another organization, you don't want to pay someone to have the same information typed over and over again.

Scope of the Topic

Links and syncs (synchronizations) between applications reduce retyping, avoid redundant data entry and deliver a variety of other benefits.

In our presentation and in these pages, we cover the concepts, advantages and drawbacks of links and syncs between software programs. We highlight a variety of desktop (premise-based) computer programs and cloud-based applications.

These materials are divided into four main topics:

Links and Syncs

Legal Software - Integrated Features and Links

Cloud Services - Integrated Features and Links

File Synchronization Technologies

We offer practical advice, precautions, limitations and strengths of various approaches and technologies for sharing and reusing information between applications.

The definitions of "integrated features," "links," and "syncs" can overlap. Here we use "integrated" to refer to features built into an application. "Links" connect one application to another, may be one-way or two-way, and tend to require a manual step to operate. "Syncs" connect separate applications or file storage locations, tend to be automatic, and are typically two-way. "Imports" and "exports" require a manual step to save data from an application to a file and another step to load the data into another application.

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He also produces, with Sharon Nelson, the monthly podcast, The Digital Edge: Lawyers and Technology. This monthly podcast covers a number of legal technology issues and can be downloaded from <http://tinyurl.com/DigitalEdgePodcast> or from iTunes. He has made hundreds of presentations on law office management, legal technology, ethics and legal business operations and has been inducted as a Fellow of the College of Law Practice Management. You can follow him on Twitter at @jimcalloway.

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Links and Syncs

No Software Does Everything

Sadly, your presenters do not believe that a "magic bullet" exists in that there is no one program that handles all of this synchronization perfectly. Too many of us focus on different areas of importance. The development of these products is still evolving, yet we doubt that one product or suite of products can be all things to all people.

There you have two broad choices:

1. An all-in-one suite with some weak components, or
2. A combination of best-of-breed programs that work together.

The line between these choices is not well-defined. Links and import/export features appear in "all-in-one" practice management applications as well as in independent, special purpose programs and cloud applications.

All-In-One Suite with Some Weak Components

For those who wish to stick with the simplicity of learning one particular package and interface, an all-in-one suite may be the best option. The loss in depth of features may be a small price to pay for an easier, simpler set of tools.

Best-of-Breed Programs Working Together

For those who place a higher value on customization and power, the best solution may be to use several independent applications together. With these combinations you rely more heavily on links and syncs. The downsides of using different programs from different companies in tandem may be more than offset by the efficiencies and greater capabilities of their larger feature sets.

Benefits of Links and Syncs

There are multiple benefits to working with links and syncs as opposed to re-entering data in multiple places.

Save Time and Increase Accuracy

The two most obvious benefits are that this saves time and increases accuracy. Admittedly as one begins working with a synchronized system, it may not feel like you are saving time as you deal with learning a new method of operation and the inevitable troubleshooting that happens when setting up a new kind of workflow. But in the long term, this should save time and, in particular, help the firm deal with "emergency" requests for something to be done within a short time frame.

Work on Multiple Devices

Another benefit is that linking and syncing makes your life much easier when you are using different devices. While you certainly could draft a long document on your smart phone, few of us would want to attempt that. There are many ways to accomplish remote access, but having strong links and syncs empowers those away from the office to be able to do so much more.

Collaboration

Collaboration is another situation where synchronized documents can shine. Both of your presenters have had experience with jointly writing papers such as this one with others in other locations and other time zones. Two people can work together fairly effectively by sending a document back and forth as an email attachment. But if there are three or more people involved in drafting, one can soon end up with several versions of a document. The process of coordinating three or four versions of the same document back into a single document is tedious and time-consuming. Although it was some years ago when it was first released, many of us remember the joy of using

Google Docs (nee Google Drive) to collaborate. There was only one version of the document even if two of us might be editing it online at the same time.

Disaster Recovery

Data and file synchronization applications do not eliminate the need for backups. You still need regular, automatic, offsite backups of your invaluable data and documents. What synchronization applications can do is make your information much more readily accessible in the event of a power outage, disk crash or more serious disaster.

By synchronizing your contacts, calendar, documents and other data to the cloud, to laptops and to mobile devices, you can continue many of your usual activities in difficult situations. That can be particularly reassuring to your clients, who may need your assistance or at least may appreciate that their attorney is well-prepared and responsive when others are temporarily out of business.

Fast, Mobile Access

Synchronization of your phone numbers, Email addresses, calendars and documents can give you fast access to what you need when you are away from the office. The ability to remotely access your practice management system from another computer can be very useful, but it may be a slow or unusable process depending on where you are and what device and connection speed is available. Synchronization can place the information immediately at your fingertips on a laptop, tablet or smartphone.

Potential Pitfalls

"Yes, we link to that."

Yeah, but how well, in which directions, and how stable is the link?

Since the synchronization features of various products are becoming more popular and are potentially game-changing, it has become almost standard to advertise and "sell" based on representations that "we link to that." All links are not created equal and a link that technically works but does not produce perfect results is potentially dangerous.

Problems with One-Way Transfers

Some links and syncs work in just one direction. If you add or change information on the destination end of the link, the original location does not receive the additions and changes. You may need to perform some work only on the source location and other work on the destination. The application may not enforce your compliance with those procedures, so you can end up with information that is missing and inconsistent.

Record Duplication, Corruption and Overwriting

Synchronization can be marvelous when it works flawlessly. If it is not set up correctly or if it is not designed to recover from potential problems that interrupt the synchronization process, the result may be record duplication, corruption or loss. Some applications do not prevent the user from performing a wrong synchronization, so human error may cause problems.

Not Everything Syncs

Links and syncs typically do not transfer 100% of the information in one application to another application. For example, you may be able to enter 30 fields of contact information in your practice management application but only 14 fields in the contact app on your smartphone. If some key information is needed in some of the other 16 fields, you need to remember to enter that information back at your computer. That raises the likelihood that not all important information will be entered.

Upgrade and Compatibility Issues

One of the most frustrating situations occurs when one has several applications working together well and an upgrade takes place which results in broken links and inoperability.

As with many other technology matters, it is generally true that delaying an upgrade for a few weeks or months is a good cautious strategy. It is also a good idea to be in contact with others who use the same configuration, whether they are in your local area or whether you are in contact with them electronically, either personally or by their public blog posts.

Attorney Fee Issues

Computers allow us to do things faster. Our clients deserve lawyers who operate in the most efficient way. Freedom from drudgery allows both lawyers and law firm staff to be sharper to handle the unique and difficult problems we encounter. In embracing more efficient technologies, we need to take into account how to bill for our more efficient services so as to avoid a potential pitfall.

But suppose a law firm invests heavily and gains immensely from a document automation project. The idea is that the information collected during the representation could be used to produce a finished document in minutes instead of several hours. The lawyers will think that is great—until they reflect on the fact that they bill their clients by the hour. So is this document any less valuable to the client because part of the production process was automated? No. If anything the opposite could be true. Automation had the potential to eliminate errors and to make it easier to update all lawyers work when there are changes in practice or law since the template for the document, which is used by all, can be updated.

In other types of businesses, an upgrade in equipment or process is most often deployed to either increase profits or remain competitive. But for a law firm committed to only billing clients by the hourly rate, any improvement that saves a lot of lawyer time can result in a reduction of fees charged. This is not always a bad thing. If a long-standing client has been making noises about considering options to reduce the fees paid to outside counsel, this may help the firm to be more competitive in pricing.

But for most law firms, it is best to consider a switch to a blended contract approach. So in the standard attorney fee contract, there can be provisions that state some language to this effect:

XYZ Firm prides itself on using modern technology to save its clients money by producing some work at a fixed fee, rather than the standard hourly rate. The following items are billed by a fixed fee:

Incorporation of a small business \$XX

Preparing annual corporate minutes.....\$XX

Drafting a covenant not to complete.....\$XX

This certainty of legal fees may be attractive to clients. But for the law firm, it means that you will receive a return on your technology investment as doing these projects should give a rate of return in excess of the lawyer's typical hourly billing rate.

The fixed fee can be set by looking at historical charges charged for this type of services and averaging them, with a discount to the client so that the fixed fee is in fact less than the average fee charged the same clients for similar services in the past. Computer automation can be good for the client and good for the lawyer at the same time. Some law firms may decide to go further and bill an entire project on a flat fee basis using this same method of calculation.

Legal Software - Integrated Features and Links

Get ready for a deeper, more technical look at what you can expect from the software you can use to perform legal work and manage a law office. In the next sections, we look at desktop or premised-based software. After that we discuss the ways that cloud-based applications integrate, synchronize and link to deliver an array of features.

Integrated features are the ones that the developers choose to build into the software. Since a single program cannot be all things to all people, the developers also create links to software produced by third-party companies.

In the following sections, we consider what features are typically integrated into practice management software and what functions are delivered by linking to third-party programs.

In the diagram, below, integrated features are enclosed in the blue oval. Each practice management program may have a different selection of integrated vs. linked functions. The diagram shows a typical division.

Legal Software: Integrated Features and Links

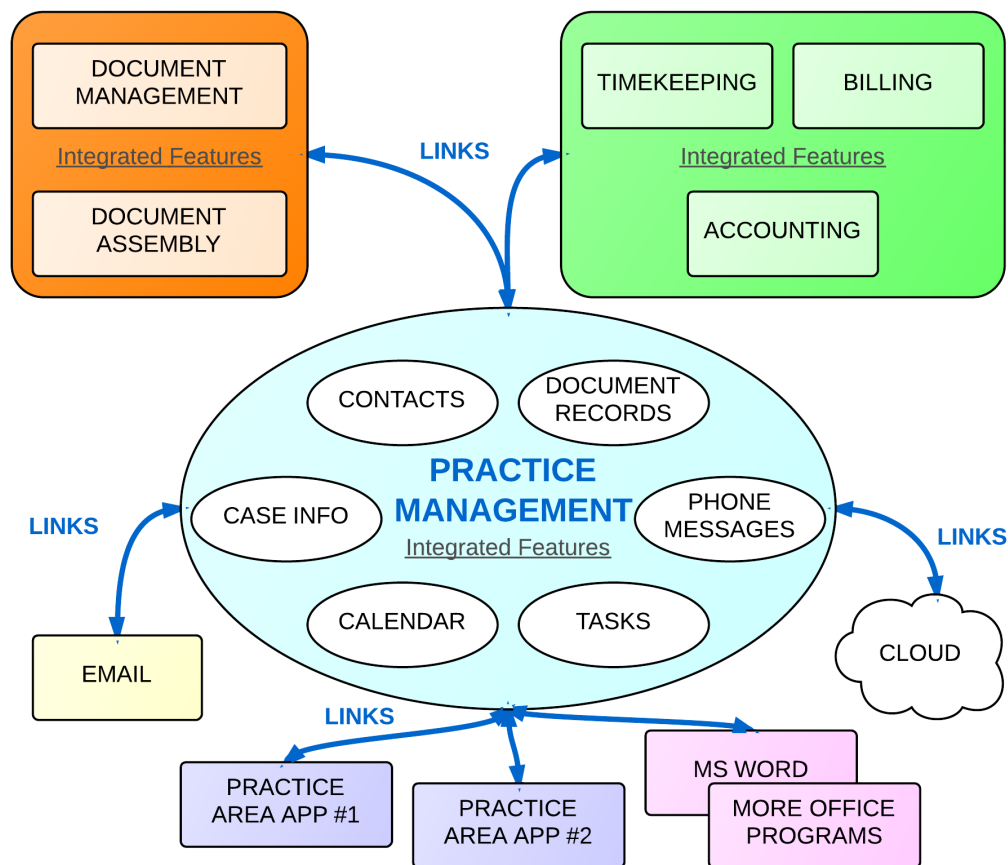


Diagram 1

Integrated Features in Practice Management Software

Each of the many desktop and server-based practice management software programs integrates a core set of law office functions. They typically include: Contact management, case information, calendars, task management, and document records. Additional integrated functions vary from software to software, such as email features, phone messages, alarms and reminders, and dashboards that summarize information. The software programs also may integrate reporting systems that allow users to create and tweak customized reports listing matters, contacts, and other information.

The integrated features in practice management software that runs locally on desktops and servers tends to be more highly customizable than their counterparts in cloud – based practice management services. They also tend to be more keystroke friendly,

allowing users to memorize hot keys that may be used more efficiently and rapidly than mouse clicks.

Most practice management software also tends to incorporate built-in timekeeping and expense recording. Many do not contain integrated functions for producing invoices and very few incorporate full-blown legal accounting.

The absence of integrated accounting features may reflect two realities:

1. QuickBooks commands a huge share of the market for small firm accounting programs. Most accountants and bookkeepers are familiar with QuickBooks, so it is a natural for small law firms to use that program.
2. Creating accounting software is complex and challenging. It requires a large investment of time and talent. Adding accounting as an extra feature to an existing application is a large undertaking.

Examples: Practice Management Software Integrated Features

Time Matters by LexisNexis

Time Matters (www.timematters.com) practice management software has long been the market leader in the small law firm marketplace. It emphasizes the following:

- Centralized case information.
- Time and expense capture.
- Extensive customization options.
- Broad set of links to other programs.
- 24/7 remote and mobile conductivity.

Time Matters contains features for entering time records from any other records within the system. In order to produce invoices, handle trust funds, and produce accounts receivable reports, Time Matters has two options:

1. Billing Matters – optional add-on software that is built into Time Matters and does not require any linking.
2. Bi-directional linking to PC Law.

Time Matters has a long list of links to other programs, including the Microsoft Office suite, WordPerfect, third-party timekeeping and accounting software, web browsers, and a variety of specialized practice area applications such as trial preparation and trial presentation software.

Amicus Attorney

Amicus Attorney (www.amicusattorney.com) competes directly with Time Matters. It emphasizes:

- Integration of all information in your law practice.
- Immediately accessible information including appointments, tasks, contacts, communications and time spent.

- Amicus Dailies module that helps answer the question, "What's going on today?"
- Universal communication system – tracking of outgoing and incoming phone calls and outgoing and incoming emails.
- Time tracking including mobile time tracking – the Amicus Time Tracker allows you to track, enter and edit time entries on your mobile phone while connected to your Amicus database over the air.
- Take Note – A legal pad for fast recording of any action item, note, fact, or task.
- Integrated knowledge management – A tool that integrates legal research into client matter files. Using the Library, you collect information resources such as memos, opinion letters, online services and internet websites.

Links from Practice Management Software to Other Programs

The desktop and local network based practice management software programs have a wide variety of links to other software programs. If you rely on specialized software to draft legal documents, produce legal forms, or manage evidence and other trial information, you are bound to find a general practice management program that can link to your specialized application.

With links to third-party applications, you are spared from the hassle and inefficiency of entering the same information in two different places. That redundant approach can cause problems when information is updated in one place but not manually updated in the other place.

Practice management software also tends to have strong links to document management systems and to generalized document assembly programs such as HotDocs. (www.hotdocs.com)

Worldox (www.worldox.com) is a powerful third-party document management system that organizes documents by matter. It indexes every word in every document to facilitate extremely fast search and retrieval of documents. Worldox integrates with many of the leading general purpose practice management software programs.

Links to Microsoft Outlook and to standard email protocols, such as POP3, SMTP and IMAP, are found in many practice management applications. It is important to investigate the details of how these email links are implemented. If your practice management program automatically downloads and removes email messages from an email server, you may find that you miss receiving important messages when you are away from the office. Most systems can be configured so that you can view emails in both places and manage their deletion and storage efficiently.

Summary - Practice Management Integrated Functions & Links

Desktop and network-based practice management software usually integrates a combination of these functions

- Contact and Client Relationship Management
- Matter and Case Information Management

- Calendar, Docket and Deadline Management
- Task Management, Delegation and Workflow
- Phone Calls and Messages
- Document Profiles

Links to other programs from practice management software may include:

- Document Management and Document Assembly
- Financial Software
- Email Software
- Practice Area Specialty Programs
- Office Software: MS Word, Excel, etc.

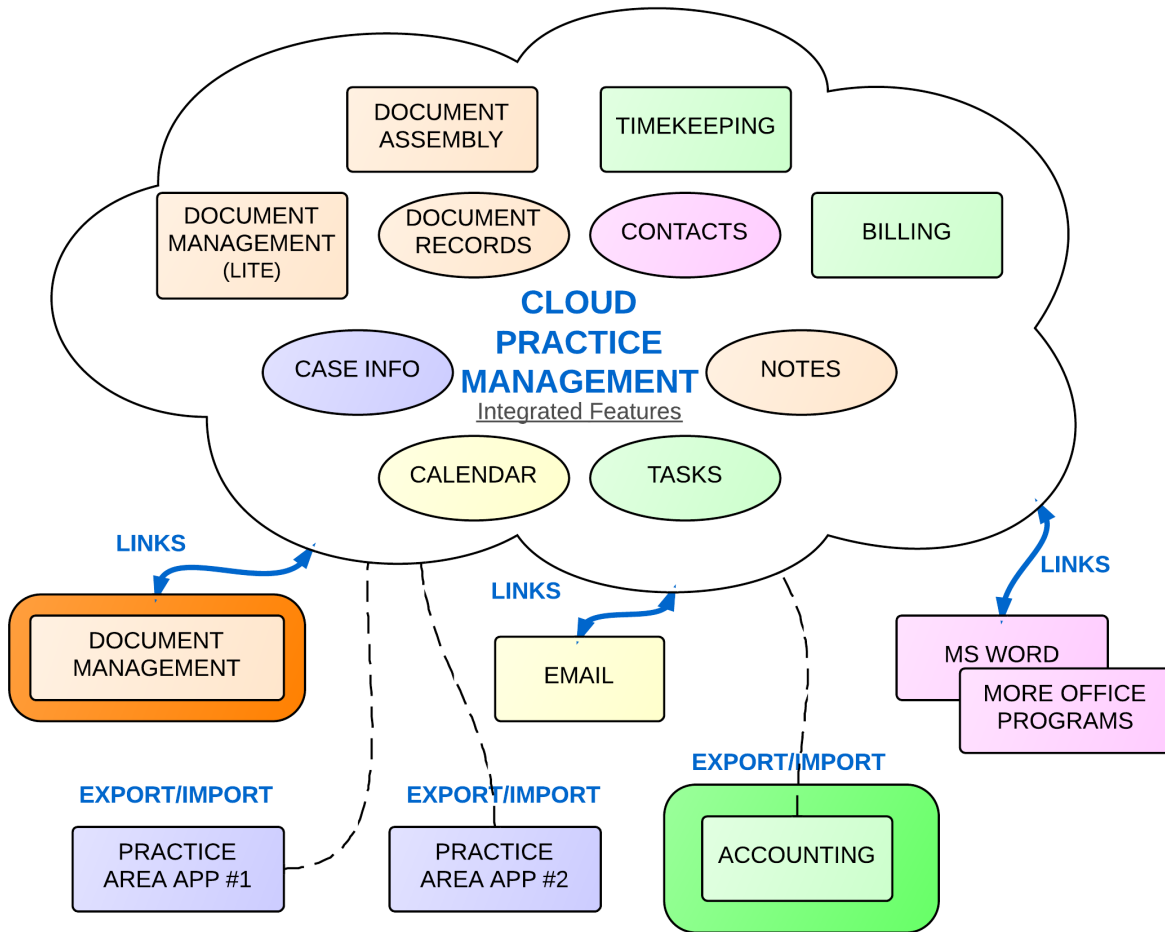
Cloud Services - Integrated Features and Links

All-in-One Cloud Practice Management

Cloud practice management services usually offer a larger mix of integrated features than their desktop and network software competition. Though their feature sets are larger and work together seamlessly, the individual features tend to be simpler, less powerful and less customizable.

The diagram, below, shows integrated features contained within the cloud outline. Links to other applications, both cloud-based and desktop-based, are shown with double-headed arrows. Export/import capabilities are shown with dotted lines.

Cloud Services: Integrated Features and Links



Cloud Services - Integrated Functions

Cloud practice management services often implement these features which may be omitted from traditional practice management software:

- Invoicing
- Accounts receivable
- Trust accounts
- Document assembly
- Document management (lite)

Cloud Services - Links to Other Programs

Cloud practice management services tend to build in fairly good links for document management. They may manage documents by:

- Working well with the file system of the Microsoft Windows or Apple operating system.
- Connecting with cloud file storage services such as Box.com, Dropbox or Google Drive.

An advantage of linking with cloud file storage services can be the ability to create and edit documents using the online service. That capability frees you from relying on word processing or other programs available on the machine you are using. More cloud practice management services link to Google Drive and Dropbox than to other cloud file storage applications.

Missing links for cloud practice management service typically are:

- Accounting applications
- Specialized practice area applications, such as bankruptcy or trial preparation

But there are exceptions. ActionStep (www.actionstep.com) includes a fully integrated accounting system and also links to cloud-based accounting from Xero (www.xero.com). Rocket Matter (www.rocketmatter.com) offers a cloud connection to QuickBooks desktop accounting software.

File Synchronization Technologies

Hot Topic! Why are so many people enthusiastic about synchronizing files?

Five reasons:

1. Get to your documents, case materials, Emails, spreadsheets, etc., fast **wherever you are**: office, home, coffee shop, conference, and jet - anywhere.
2. Protect all your files from **disasters** by keeping redundant copies in multiple locations.
3. Make changes in one place and all your redundant **copies are updated** within minutes or at least within hours.
4. **Share your files** securely with colleagues, clients, experts - anyone you choose.
5. Synchronization is **cheap**!

Don't rewrite something you know *you have already written!* If you have your files right at hand wherever you are working, you can reuse what you have written before.

Don't rewrite a document because *you lost it!* Make sure whatever you write is stored in multiple places so you will never have to recreate it from scratch.

What you need is a system that makes your files available where you are and that allows you to find what you need fast. For that, you need two technologies:

- File synchronization to serve up your files and to sync your new writing back to the office or other location
- File finding tools to rapidly locate what you need.

How Does File Synchronization Work?

File synchronization services run an application on each of the machines you want to synchronize. Any combination of your server, office computers, laptops, smartphones and tablets may be supported.

When a file is added or changed on one machine, it is added or changed on each of the other machines, which may include cloud storage. The additions and changes may occur in real time or on a schedule when the machines are connected through a local network or the Internet. For machines that are off the network, the updates may be when they are reconnected to the network or the Internet. The updating may be automatic, on a schedule, or on demand.

Synchronization may also take place through the air (WiFi, Bluetooth, cellular data) or via a cable connection between two machines.

Each synchronization system must handle a number of potential problems. Some of the problems are:

- Copying open or locked files.
- Handling file conflicts that can result when the same file is changed by different users in between synchronizations.
- Managing deleted files and old versions.
- Controlling access to files.
- Recovering if the network connection is broken in the middle of synchronization.

When you compare and evaluate different synchronization products, you need to consider how each one handles the problems that may affect your use.

Huge Table: Comparison Synchronization Software

Included as an appendix to this document is a huge table comparing many file synchronization software products and some cloud synchronization services:

Comparison of Sync Software.pdf

Caveat: The table should be considered a *starting point only* for your research and evaluation. The table is necessarily *incomplete*. Vendors do not uniformly and consistently list the features of their products. The table is an effort to collect most of the features of the listed products, but is *not comprehensive*. Any errors or omissions are unintentional. By the time you read this, the table will be *out of date*. Technology moves fast. Please refer to each product's website for more complete, up-to-date information.

Different File Synchronization Technologies

File synchronization technologies fall into three general categories:

1. **Cloud File Storage and Synchronization** - Your files are copied to cloud storage. From there they are synchronized between your cloud storage space and your computers and other devices.
2. **Local File Synchronization** - Your files are copied between your computers and devices in the same room or on the same local network.
3. **Remote File Synchronization** - Your files are copied between your computers and other devices across the Internet. They are not stored in the cloud.

Each of these technologies has its own set of advantages and drawbacks. With each technology, each product varies in quality, features and security.

Cloud File Storage and Synchronization

The buzz lately is about storing, synchronizing and sharing files in the cloud. Examples of cloud file storage and synchronization services are:

- Box.com
- Dropbox
- Egnyte
- Google Drive
- Microsoft SkyDrive
- SugarSync

Cloud Sync Advantages

Chief advantages of cloud sync services include:

- + Ease of setup and use
- + Automatic operation
- + Low or no cost
- + Compatibility with other cloud services
- + Added features such as file sharing
- + Updating and maintenance automatically by vendor

Cloud Sync Drawbacks

Chief drawbacks of cloud sync services include:

- Occasional security breaches at some vendors
- Lack of administration and control features for multiple users
- Lack of detailed information about features and sync characteristics
- Inability to sync files from existing folders

Local File Synchronization

Local file synchronization serves a number of purposes:

- Allow use away from the office by syncing files from a server to a laptop
- Make files immediately available in the event of a server crash or network outage by syncing files from an internal server drive to an external hard drive
- Synchronize files on a computer workstation to the server to ensure they are backed up regularly

A variety of free and inexpensive software applications are available for local files synchronization. They offer widely varying degrees of power, flexibility, security and automation.

See the Appendix, *Comparison of Sync Software*, for details on many software products and the leading cloud file storage and sync services.

Remote File Synchronization

Remote file synchronization is used here to refer to systems that synchronize files between machines across the Internet without using a cloud synchronization service.

A distinction is that you pay a one-time purchase price for the synchronization software and you set up the destinations for the synchronization of files.

All of your synchronizations may go directly to machines you own or you may rent storage space on a cloud server (FTP server) that acts as a way station. You control how that space is used and managed.

Remote file synchronization may be performed by the same software products used for local file synchronization. The software must support one or more file transfer protocols such as FTP, SFTP, FTPS, and WebDAV or use a virtual private network (VPN).

Chief advantages of remote file synchronization:

- Control - You configure and manage the software and hardware
- Cost - You do not pay monthly subscriptions based on space usage but rather purchase software and machines outright.

Chief drawbacks of remote file synchronization:

- Control - You do not gain the benefit of outsourcing the configuration and management of your system to a large, professional vendor
- Maintenance - You have to ensure that the system is maintained and the software is updated as needed

Features of File Synchronization Software

Cost

Many synchronization software programs and synchronization services make a free version available. This accompanying table, *Comparison of Sync Software*, lists some programs that are 100 percent freeware and lists some of the free versions of commercial programs.

Not all free versions are listed. Most have such limited feature sets or such strict limitations on numbers or size of files that they are not useful for real work. Such free versions can introduce you to the experience of using the program, but you need their paid versions for day-to-day use.

One-Way and Bi-Directional

Most synchronization software and services perform bi-directional synchronization by default. Some products also support one-way synchronization or backup. Within these classifications, the applications may work in various ways. It is important to consider the versioning and retention features of a synchronization application. Some retain old versions for a limited period of time, some indefinitely (which can lead to large consumption of disk space), and some maintain only the most recent version of a file in all locations.

Conflict Resolution

Conflict resolution is a feature of *all file sync software* so it is not listed in the accompanying table, *Comparison of Sync Software*.

Conflict resolution options determine what happens if the same file is changed in different locations between the times that files are synchronized. Options may include:

- Newer file overwrites older file.
- Source file overwrites destination file.
- Both files are preserved and older file is renamed by appending a date or number to the file name.
- Both files are preserved using a versioning feature controlled by the software.

How file conflicts are resolved may be the most important factor for you to understand about your synchronization software. For example, you make changes to a legal memorandum on your laptop while not connected to the Internet. Later the same day, your partner is reading the memorandum at the office, notices a typographical error, and corrects it. If your synchronization software or service overwrites the older file with the newer file, all your changes may be lost. If there is no version system or you don't understand it, you could end up rewriting hours of work.

Transfer Only New Blocks

File synchronization can occur by transferring entire files, or by transferring just the blocks of individual files which have changed. (Blocks are the many small pieces into

which files are divided by the operating system.) Transferring only new and changed blocks very significantly reduces the amount of data that needs to be transferred across a network or the Internet. This significantly influences the amount of time required to synchronize files.

Fast Scan or Tracking Database

Applications that use a tracking database and fast scanning can complete a synchronization process much more rapidly than applications which do not have this feature. Fast synchronization reduces the sluggishness that can otherwise occur during the synchronization process.

Network Drive Backup Source

Some applications or services do not have the capability of synchronizing files located on other computers on a network.

Network Drive Destination

Some synchronization applications cannot synchronize files to another computer's hard drive across a network, but rather can only synchronize to another computer that is also running the synchronization application.

Resume Interrupted Transfer

It is possible for a synchronization process to be interrupted or halted as a result of a problem on the Internet, or a problem with the connection of a computer to another computer or to the Internet. Different applications have varying degrees of resiliency in responding to an interrupted transfer. The best applications pick up where they left off.

Encryption and Compression

Encryption is important to protect synchronized files as they are transferred across the public Internet. Some applications also support encryption of synchronized files in place on a far drive or machine. Compression of synchronized files before transfer is the rule rather than the exception with synchronization applications. Compression reduces the amount of time that a machine must spend processing and transferring new and changed data.

Verification

Applications that support verification perform a check of files after they have been transferred across a network.

Renamed File Detection

An unusual feature in synchronization applications, renamed file detection, operates to identify files that have only had their name changed rather than their contents. Without renamed file detection, a renamed file is treated as a new file and the file with the old name may be deleted because it appears to have disappeared from the source machine.

Scheduling Options

Many synchronization applications perform continuous or realtime synchronization. They may or may not have the ability to set a schedule, such as synchronization every half hour or once per day.

Filtering – Select and Exclude

Filtering is a key feature for many users of synchronization software. It permits the user to decide which files or folders should be synchronized and which ones should not. This feature allows the user to drastically reduce the size and, therefore, the speed of synchronization operations by excluding large files or folders which would slow down the process and unnecessarily consume space.

Copy File and Folder Dates

Almost all synchronization applications accurately copy file dates when synchronizing files. An exception can be when files are synchronized to an FTP server. Sync applications need to use special technologies in order to support accurate preservation of file and folder dates. Often the folder date of a folder created on a far device will be recorded as the date the folder was first copied across the network, rather than the original date of the folder itself.

Deleted Files – Keep or Remove

In order to prevent the accidental, permanent loss of deleted files on all synchronized machines, some applications allow the user to select an option to keep all deleted files sometimes in a Deleted Files folder, or remove them entirely during synchronization.

Forum

The application or service has a user forum in which questions are asked and answered. A forum can be an excellent place to identify whether a particular application or service is prone to problems, or whether any problems are rapidly resolved by the developers and the user community.

Throttling – Bandwidth Limiting

If a synchronization application is free to use all available Internet bandwidth when uploading and downloading files, it can seriously impact the performance of Internet use by other users. Throttling or bandwidth limiting allows the user to configure a limit or ceiling on the amount of available upload or download capacity that the synchronization application is permitted to consume.

FTP – SFTP Support

FTP and SFTP protocols are standard methods of file transfer across the Internet. The SFTP protocol is a secure method that encrypts the files in transit. These protocols can be used with inexpensive FTP server space available for monthly subscription from many Internet service providers.

Realtime or Continuous Synchronization

Whenever a file is added, deleted or changed, a synchronization application with a realtime capability will immediately transfer the changes across the network. This feature cuts down on the likelihood that a single file might be changed by different users in different locations resulting in a file conflict.

Versioning by Date

When the same file is modified multiple times, the versioning by date feature keeps track of each version of the file and associates a date with that file. Many synchronization applications simply transfer a new version of a file across the network and overwrite the older version of that file on the other machines.

Retention

A retention feature allows the user to specify how long or how many versions of a changed file should be retained.

Email Notification

The email notification option can be enabled in order to alert a user to failed synchronization operations, or even to successful synchronization operations.

Runs as a Service

Computer services can run on a computer even while the user is logged off a computer or has locked the computer screen without powering the machine down. File synchronization can take place under these circumstances if the application can run as a service.

Copy Open Files and Locked Files

Files that are currently being edited or are in use typically cannot be copied. However, some synchronization applications are designed to synchronize copies of documents that are in use or locked.

Options for Naming Destination Folders

This unusual feature allows each synchronization to create destination folders with names that contain serial numbers or dates and times.

Automatically Run on Drive Connection

If a machine, such as a laptop, is frequently off the network, with this feature it can be configured to automatically synchronize its files when it is reconnected to the network.

Tech Support

The quality of technical support can be extremely important if you encounter a problem with the setup or synchronization of files. Some vendors provide excellent forums and technical information on their web sites. Others offer only simple documentations and little or no technical support.

Features of Cloud Storage and Synchronization Services

The leaders in cloud storage and synchronization – Dropbox, Google Drive, Microsoft SkyDrive, focus on *keeping things simple*. They target the mass market, keeping pricing low and designing features so that they require little or no assistance from technical support.

As a result of their focus on simplicity, the mass-marketed services offer less user control over how features are customized and place more limitations on the services.

For example, the leading three services do not allow you to synchronize files from their original locations in the folder structure on your hard drive. Instead, you need to copy or drag and drop files and folders from their original locations *into a special folder* created by the cloud storage service on your hard drive.

This significant limitation on the flexibility of most cloud storage services can be reduced by using the mapped drive feature that some services offer. You map a drive letter to your file storage area in the cloud. Your files and documents are located in the cloud, but you can access them using the new drive letter and the folders for your documents and files in the cloud.

A drawback of the mapped drive feature can be performance. Because you are pulling your files down from the cloud across the Internet and saving them back up to the cloud, the process can be noticeably slower depending on the speed of your Internet connection.

Cloud File Synchronization

Cloud storage services typically allow you to synchronize your files up in the cloud to multiple machines, including desktop computers, servers, tablets, and smart phones. When you work with files that have been synchronized to your device, you have the benefit of fast performance; however, you also need to be aware of the characteristics of the synchronization process for handling file conflicts. A file conflict arises when, unplugged from the Internet, you change a local copy of a file and someone else changes the copy of the same file up in the cloud. Most services offer versioning so that both versions of the document are preserved. You need to know how that system works so that you can find the version you need.

Secure File Sharing

The term "file sharing" may have a bad name as a result of controversies around sharing commercial songs and movies. But file sharing in the context of a law practice can be a powerful tool for providing convenient, fast client service.

The secure file sharing feature offered by many cloud file storage services allows you to exchange files with clients, experts and other people by sending secure links to files or by providing restricted access to particular files.

Each of these two approaches to file sharing has drawbacks. When you send a secure link, you send it using an insecure medium: email. Though the link to the document is complex enough that it cannot be guessed by an outsider, the email itself is open to

being intercepted as it makes its way across the Internet. The other method, giving restricted access to documents in your cloud, requires that the recipient either log into your cloud with the user name and password or already have an account with the same vendor. Therefore, if you wish to share access to a file, your recipient already needs to have a login or an account ahead of time.

Conclusion

Do Your Homework

Get specific in your research. If you need to use two separate programs together, identify in detail how they are synchronized or linked. What are the requirements and limitations? What are the real benefits? What is the history of problems or lack of problems for the two products working together?

Read the discussion forums where users ask questions. Talk to another firm that has used the combination you are considering. Speak with a consultant who has experience with the products in question.

Take Advantage of Links and Syncs

Syncs and links can tie together the tools that will work hard for you making your job easier. By choosing effective combinations of the best software and cloud services, you can be more responsive, be more efficient and actually have more fun doing your work.

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Appendix 1 - Compare Cloud Practice Management Services

COMPARE CLOUD PRACTICE MANAGEMENT SERVICES	PRICING Per Mo.	ACCOUNTING FEATURES	Accounts Receivable	Trust Funds	Chart of Accounts	Bank Accts, Credit Card Accts
ActionStep	\$59.95/user Free for external, limited users (Clients, Accountant, Experts ...)		Yes	Yes	Yes	Yes
Clio	\$49/Atty \$24.95/Staff		Yes	Yes	No	No
MyCase	\$39/Atty \$29/Staff		Yes	Yes	No	No
Rocket Matter	\$59.99/user Qtrly: -10% Yrly: -15%		Yes	Yes	No	No
Amicus Cloud	\$35/user		Yes	Yes	No	No

COMPARE CLOUD PRACTICE MANAGEMENT SERVICES	Import Banking Transactions	Accounting Reports	QuickBooks Export	UTILITIES & MOBILITY FEATURES	Import Records	Export Records
ActionStep	Yes when linked to Xero (\$29/mo.)	Yes	No		Contacts in groups by Type, Matters, Links: Matters & Contacts, Calendar	Export any list to Excel .csv: Matters, Contacts, Tasks, etc.
Clio	No	No	Invoice, Operating Account, Trust Account		Contacts, Matters, Calendar, Notes, Tasks, Time Records	Export all records in Excel .csv Export Bills, Bank txns in .csv and .iff
MyCase	No	No	No		Contacts, Companies, Matters	Yes All records in CSV format.
Rocket Matter	No	No	Better: Aggregate data into QB via cloud		Contacts, Matters, Billing, more	Contacts, Matters, Billing, no custom fields
Amicus Cloud	No	No	Yes		Calendars, Contacts, Tasks and more	Time & Exp. Sync to Outlook+M1

COMPARE CLOUD PRACTICE MANAGEMENT SERVICES	Backup and Download	Exchange Sync	Synchronize Calendar	Synchronize Contacts	Smartphone Web Access (Mobile pages)	Smartphone App (offline)
ActionStep	Yes Data and Documents in CSV format. Backup all data in one batch.	No	2-way: Google, Outlook (via Google)	No	Yes 2-way Calendar, Contacts, Matters, Time Entry	No
Clio	Yes Data and Documents	Sync only with local Outlook copy - OST - Cached Exchange Mode	2-way sync with Google Apps, Outlook.	2-way sync with Google Apps, Outlook.	Yes	iPhone / iPad Android
MyCase	Yes All records in CSV format. ZIP file with 17 files.	No	2-way: Google, Outlook (PC), Apple iCal	No	No	No
Rocket Matter	No backup feature. Use multiple exports.	Sync with Outlook. Sync Exchange Email	iPhone / iPad Google Apps Android	iPhone/iPad	Use iPhone app	iPhone / iPad
Amicus Cloud	Yes	Yes + Outlook Integration	iPhone / iPad Android	iPhone / iPad Android	iPhone / iPad Android	No

Disclaimer
The table is a starting point only for your research and evaluation. It is necessarily incomplete. Vendors do not uniformly and consistently list the features of their products. The table collects most of the features of the listed products, but is not comprehensive. Any errors or omissions are unintentional. By the time you read this, the table will be out of date. Technology moves fast. Please refer to each product's website for more complete, up-to-date information. February 2013
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Appendix 2 - Comparison of Sync Software

Comparison of Sync Software	4-Star		5-Star	
Product	ViceVersa Pro	ViceVersa (Svr)	GoodSync	Livedrive
Important Features				
Cost	59.95 + 24.95	129/189 + 94	29.95, 2nd 9.95	59.95 reseller
Limitations			Server: \$995	
One-way and bi-directional	Yes		Yes	
Differential and incremental			Differential (1-way)	30 versions
Transfer only new blocks	No		Yes - Ver. 9	
Fast scan or tracking database	Yes		Yes	
Network drive source (map drive)	Yes		Yes	
Network drive destination	Yes		Yes	No
Resume transfer of partial file	No, but Yes job		No, but Yes job	Yes
Encryption and compression	Yes - file by file		Yes	
Verification				
Renamed file detection			Yes	
Scheduling options	Yes		Yes, many	Continuous
Filtering: select and exclude	Yes		Yes	Yes?
Copy file and folder dates	Yes		YES FTP	
Deleted files - keep or remove	Yes		Yes	30 days
Forum	Yes		NO	Yes
Throttling / bandwidth limiting			Yes	No*
FTP / SFTP support	FTP, date issue		Yes / Yes	
Optional Features				
Continuous sync - realtime	Yes		Yes	Yes
Versioning by date	Yes - archive		Last 2 versions	30 versions
Retention: purge versions/days				30 versions
E-mail notification	Yes		Yes	
Runs as a service	Yes (opt. VVEngine)		Yes - Ver. 9	
Copy open/locked files - VSS	Yes		Yes	
Naming destination folders			Enterprise only	
Auto run on drive connection	Yes, + net dest.		Yes	
Tech Support				Unresponsive
Special features	Auto encryp/de-		WebDAV, SFTP	Server backup
	Multi source, targ.	Replicate SQL Server		Web portal
Webpages	tgrmn.com	tgrmn.com	goodsync.com	livedrive.com

Comparison of Sync Software

		4-Star	4-Star	5-Star
Product	BitReplica	FreeFileSync	SyncBack Free	SyncBack SE
		Open Source		
Important Features				
Cost	FREE	FREE	FREE cmdl OK	29.95 / Pro 54.95
Limitations				\$24 ea/10
One-way and bi-directional	Yes	Yes		Yes
Differential and incremental	Yes	Yes? dated folders	* Ignore if older	Yes
Transfer only new blocks	NO	Yes		
Fast scan or tracking database	?	Yes	No	Yes
Network drive source (map drive)	Yes	Yes	Yes	Yes
Network drive destination	Yes	Yes	Yes	Yes
Resume transfer of partial file	?		?	
Encryption and compression	NO / Yes	No	Yes	Yes
Verification		Yes*		
Renamed file detection		No		
Scheduling options	Yes	Yes	Yes	Yes
Filtering: select and exclude	Yes	Yes	Yes	Yes
Copy file and folder dates	Yes	Yes (not FTP)		
Deleted files - keep or remove		Yes		
Forum		Yes		Yes
Throttling / bandwidth limiting			NO	Yes
FTP / SFTP support	NO		Yes-FTP	Yes (Pro:SFTP)
Optional Features				
Continuous sync - realtime		Yes BUT hard	No	Yes
Versioning by date		Yes		
Retention: purge versions/days				
E-mail notification			Yes	Yes
Runs as a service	Yes	Yes BUT hard		Yes - set backgrd
Copy open/locked files - VSS	NO	Yes	No	Yes
Naming destination folders		Yes	No	Yes
Auto run on drive connection		Yes		Yes
Tech Support				
Special features	Sync	Drag/drop to select		
	No file deletion	Portable version		
Webpages	BitReplica.com	sourceforge.com	2brightsparks.com	2brightsparks.com

Comparison of Sync Software

	3-Star	4-Star	5 Star	
Product	Allway Sync	Allway Sync	Synccovery	Synccovery
	Free	Pro	Standard	Pro
Important Features		FTP support		
Cost		19.95 / 2nd: 9.95	39.95	59.95, 5x40
Limitations	<40,000 files scan			
One-way and bi-directional		Yes	Yes	Yes
Differential and incremental			Yes	Yes
Transfer only new blocks		No	Yes	YES + part update
Fast scan or tracking database		YES	Yes	Yes
Network drive source (map drive)		Yes		
Network drive destination		Yes	Yes	Yes
Resume transfer of partial file		NO	YES	YES
Encryption and compression		Yes - by folder opt	Yes	Yes
Verification				
Renamed file detection			Yes + Moved files	Yes + Moved files
Scheduling options		Yes	Yes	Yes
Filtering: select and exclude		Yes	Yes	Yes
Copy file and folder dates		Yes	Yes	Yes
Deleted files - keep or remove		Yes	Yes	Yes
Forum		NO	Yes	Yes
Throttling / bandwidth limiting		NO	Yes: speed limit	Yes: speed limit
FTP / SFTP support		Yes	NO	Yes, date issue
Optional Features				
Continuous sync - realtime		Yes	No	Yes
Versioning by date				
Retention: purge versions/days				
E-mail notification				
Runs as a service			No	Yes
Copy open/locked files - VSS		NO	Yes	Yes
Naming destination folders				
Auto run on drive connection				
Tech Support				
Special features				
Webpages	allwaysync.com	allwaysync.com	synccovery.com	synccovery.com

Comparison of Sync Software

Product	Beyond Sync	Beyond Sync	Create	DirSyncPro
	Standard	Pro	Synchronicity	
Important Features				
Cost	29.95	49.95	FREE Open Src	FREE
Limitations	4 root folder tasks			
One-way and bi-directional	Yes	Yes	Yes	Yes
Differential and incremental			Incremental	
Transfer only new blocks			NO	
Fast scan or tracking database	YES	YES		NO
Network drive source (map drive)			Yes	
Network drive destination	Yes	Yes	Yes	NO
Resume transfer of partial file			Resume skipped	NO
Encryption and compression	Yes	Yes		NO
Verification	Yes	Yes		Yes
Renamed file detection	Yes	Yes		
Scheduling options	Yes	Yes	Some	Yes
Filtering: select and exclude	Yes	Yes	Yes	
Copy file and folder dates				
Deleted files - keep or remove				
Forum	NO	NO		NO
Throttling / bandwidth limiting				
FTP / SFTP support				
Optional Features				
Continuous sync - realtime	Yes	Yes	NO	NO
Versioning by date			Yes	
Retention: purge versions/days				
E-mail notification				
Runs as a service	Yes	Yes		
Copy open/locked files - VSS				
Naming destination folders				
Auto run on drive connection				
Tech Support				
Special features	Fast			Pre-sync backup
	Access to PCs			Java-based
Webpages	fevosoft.com	fevosoft.com	sourceforge.net	dirsincpro.org

Comparison of Sync Software

Product	Synkron	BestSync	BestSync Free
		Google, Amazon	
Important Features			
Cost	FREE	\$19.95/38/78	FREE
Limitations	Limited features		
One-way and bi-directional		Yes	Yes
Differential and incremental		Yes	Yes
Transfer only new blocks			
Fast scan or tracking database			
Network drive source (map drive)		Yes	No
Network drive destination		Yes	No
Resume transfer of partial file		Yes	No
Encryption and compression	No No	Yes	NO
Verification		Yes	Yes
Renamed file detection			
Scheduling options		Yes	Yes
Filtering: select and exclude			
Copy file and folder dates		Yes	Yes
Deleted files - keep or remove		Yes	No
Forum		Yes	Yes
Throttling / bandwidth limiting		Yes	No
FTP / SFTP support		Yes w/timestamp	No
Optional Features			
Continuous sync - realtime		Yes	No
Versioning by date		Yes	Yes
Retention: purge versions/days			
E-mail notification		Yes	No
Runs as a service		Yes	No
Copy open/locked files - VSS		Yes	No
Naming destination folders			
Auto run on drive connection		Yes	No
Tech Support			
Special features		Resolve conflicts	Resolve conflicts
		Fast!	Fast!
Webpages	sourceforge.net	risefly.com	risefly.com

Appendix 3 - Comparison of Sync Services

Comparison of Sync Services	SYNC SERVICES			
Product	IDriveSync	Syncplicity	Dropbox	Box.com
			PM links	PM links
Important Features	1 Folder only	Edit files online	100/200/500+ GB	25/50/1000 GB
Cost	4.95/150GB 14.95/500GB	\$45/3users/50GB	\$10/\$20/\$50+/mo.	\$10/\$20/\$45+/mo.
Limitations	Not geo-redund	No USB/mapped drive		File size: 1-2GB
One-way and bi-directional	Bi-directional	Bi-directional	Bi-directional	Bi-directional
Differential and incremental	30 Versions			
Transfer only new blocks			Yes	
Fast scan or tracking database				
Network drive source (map drive)	No mapped	Yes		
Network drive destination		Yes		
Resume transfer of partial file			Yes	Yes
Encryption and compression	Yes	Yes	Yes	Yes
Verification				
Renamed file detection				
Scheduling options	Auto or 1-time			
Filtering: select and exclude				
Copy file and folder dates				
Deleted files - keep or remove	?	Yes	1 mo./unltd \$40/yr	
Forum	No			
Throttling / bandwidth limiting	?		Yes	
FTP / SFTP support				
Optional Features				
Continuous sync - realtime		Yes	Yes	No
Versioning by date		Yes		Yes
Retention: purge versions/days		Yes		
E-mail notification				
Runs as a service				
Copy open/locked files - VSS	Yes			
Naming destination folders				
Auto run on drive connection	Negative reviews			
Tech Support	Instant Msg only			
Special features	Web required	offline/online	Share files	Share files
	Unltd PCs	any folders	Mobile features	Google Apps
Webpages	IDriveSync.com	syncplicity.com	dropbox.com	box.com

Comparison of Sync Services

Product	Egnyte	Google Drive	MS SkyDrive	SugarSync
Important Features	Select any folders	Edit files in cloud	SharePoint integration	Select any folders
	150/1000/2000 GB	20/50/200+ GB	20/50/100	30/60/100/250+
Cost	\$25/\$70/\$130 /mo.	\$4/\$7/\$17+ /mo.	\$10/\$25/\$50 /mo.	\$5/\$10/\$15/\$25+
Limitations				
One-way and bi-directional	Yes	Yes		Yes
Differential and incremental				
Transfer only new blocks	Yes			
Fast scan or tracking database				
Network drive source (map drive)	Yes			Yes
Network drive destination	Yes			Yes
Resume transfer of partial file	Yes	Yes	Yes	Yes
Encryption and compression	Yes	Yes	Yes	Yes
Verification				
Renamed file detection				
Scheduling options	Yes			
Filtering: select and exclude		Google folder only		
Copy file and folder dates	Yes			
Deleted files - keep or remove				
Forum				
Throttling / bandwidth limiting	Yes			
FTP / SFTP support	Yes			
Optional Features				
Continuous sync - realtime	Yes	Yes	Yes	Yes
Versioning by date	Yes	30 days, opt. more	No	Yes
Retention: purge versions/days		Save forever option		
E-mail notification	Yes			
Runs as a service	Yes			
Copy open/locked files - VSS	Yes			
Naming destination folders				
Auto run on drive connection				
Tech Support				
Special features	Hybrid Cloud	Mobile access		Any folder
	Sharing	Sharing	Sharing	Sharing
Webpages	egnyte.com	google.com/drive	skydrive.com	sugarsync.com

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By the time you read this, the table will be out of date. Technology moves fast. Please refer to each product's website for more complete, up-to-date information.

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